

Trees

Introduction to Trees

Definition 1: A *tree* is a connected undirected graph with no simple circuits.

Example: Which of the graphs shown in Figure 2 are trees?

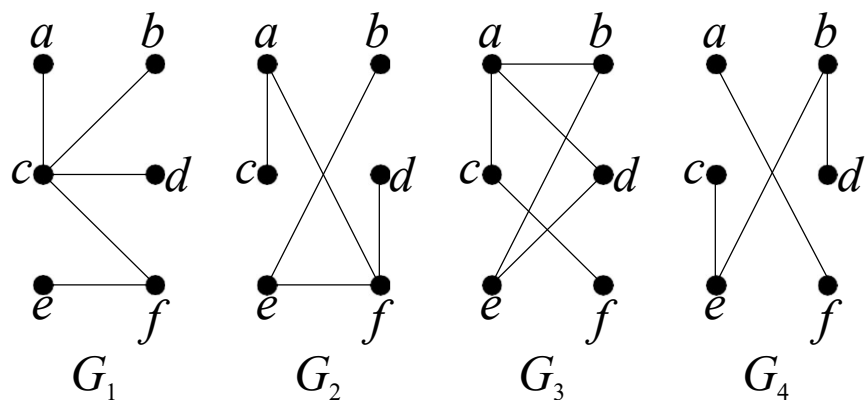


Figure: Examples of Trees and Graphs That Are Not Trees.

Solution:

G_1 and G_2 are trees, since both are connected graphs with no simple circuits.

G_3 is not a tree because e, b, a, d, e is a simple circuit in this graph.

G_4 is not a tree since it is not connected.

Theorem 1: An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

Definition 2: A *rooted tree* is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

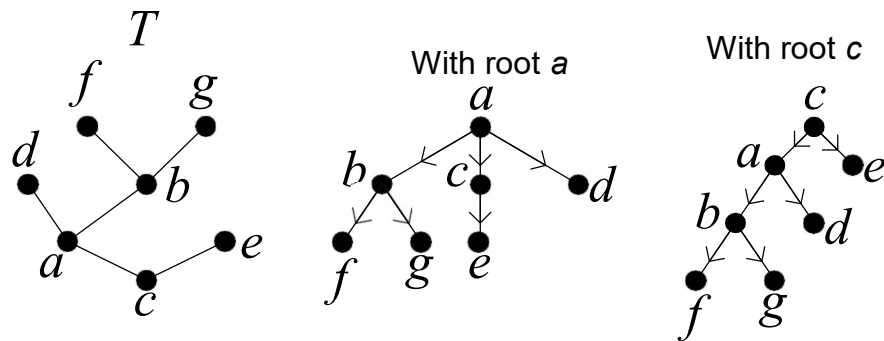


Figure: A tree and Rooted Trees Formed by Designating Two Roots.

We usually draw a rooted tree with its root at the top of the graph. The arrows indicating the directions of the edges in a rooted tree can be omitted, since the choice of root determines the direction of the edges.

Suppose that T is a rooted tree.

If v is a vertex in T other than the root, the **parent** of v is the unique vertex u such that there is a directed edge from u to v .

When u is the parent of v , v is called a **child** of u .

Vertices with the same parent are called **siblings**.

The **ancestors** of a vertex other than the root are the vertices in the path from the root to this vertex, excluding the vertex itself and including the root.

The **descendants** of a vertex v are those vertices that have v as an ancestor.

A vertex of a tree is called a **leaf** if it has no children.

Vertices that have children are called **internal vertices**.

If a is a vertex in a tree, the **subtree** with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

Example: In the rooted tree T (with root a) shown in Figure 5, find the parent of c , the children of g , the siblings of h , all ancestors of e , all descendants of b , all internal vertices, and all leaves. What is the subtree rooted at g ?

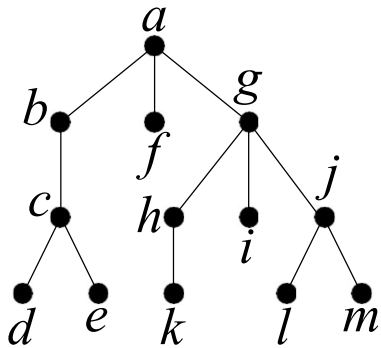


Figure: A Rooted Tree H .

The subtree rooted at g is shown in Figure 6.

Solution:

The parent of c is b .

The children of g are h , i , and j .

The siblings of h are i and j .

The ancestors of e are c , b , and a .

The descendants of b are c , d , and e .

The internal vertices are a , b , c , g , h , and j .

The leaves are d , e , f , i , k , l , and m .

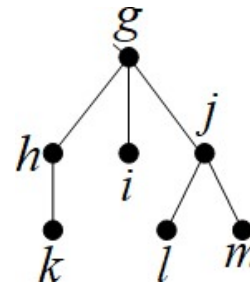


Figure 6. The Subtree Rooted at g .

Definition 3: A rooted tree is called an m -ary tree if every internal vertex has no more than m children. The tree is called a *full m -ary tree* if every internal vertex has exactly m children. An m -ary tree with $m = 2$ is called a *binary tree*.

Example:

Are the rooted trees in Figure 7 full m -ary trees for some positive integer m ?

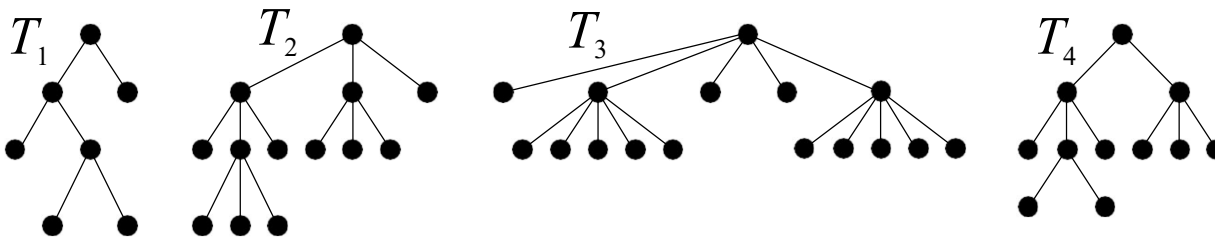


Figure: Four Rooted Trees.

Solution:

T_1 is a full binary tree.

T_2 is a full 3-ary tree.

T_3 is full 5-ary tree.

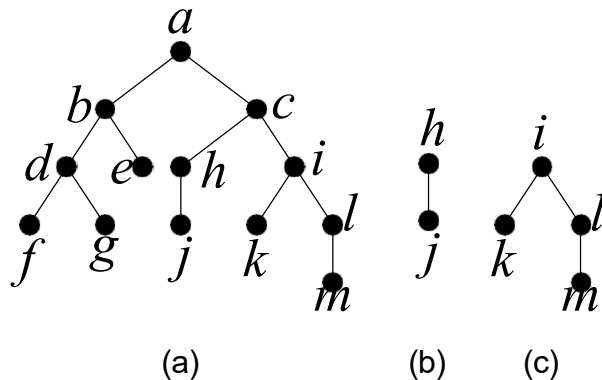
T_4 is not a full m -ary tree for any m since some of its internal vertices have two children and others have three children.

An **ordered rooted tree** is a rooted tree where the children of each internal vertex are ordered. Ordered rooted trees are drawn so that the children of each internal vertex are shown in ordered from left to right.

In an ordered binary tree (usually called just a **binary tree**), if an internal vertex has two children, the first child is called the **left child** and the second child is called the **right child**.

The tree rooted at the left child of a vertex is called the **left subtree** of this vertex, and the tree rooted at the right child of a vertex is called the **right subtree** of the vertex.

Example: What are the left and right children of d in the binary tree T shown in Figure 8(a)? What are the left and right subtrees of c ?



Solution:

The left child of d is f and the right child is g .

We show the left and right subtrees of c in Figures 8(b) and 8(c), respectively.

Figure: A Binary Tree T and Left and Right Subtrees of Vertex c .

Properties of Trees

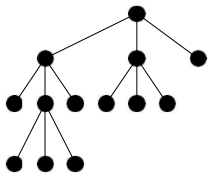
Theorem 2: A tree with n vertices has $n - 1$ edges.

Theorem 3: A full m -ary tree with i internal vertices contains $n = mi + 1$ vertices.

Theorem 4: A full m -ary tree with

- (i) n vertices has $i = (n - 1)/m$ internal vertices and $l = [(m - 1)n + 1]/m$ leaves,
- (ii) i internal vertices has $n = mi + 1$ vertices and $l = (m - 1)i + 1$ leaves,
- (iii) l leaves has $n = (ml - 1)/(m - 1)$ vertices and $i = (l - 1)/(m - 1)$ internal vertices.

Example:



The above tree is a full 3-ary tree, that is, $m = 3$.

The number of vertices is $n = 13$.

The number of internal vertices is $i = 4$.

The number of leaves is $l = 9$.

From Theorem 2,

The number of edges is $e = (n - 1) = 13 - 1 = 12$.

From Theorem 4,

(i) $i = (n - 1)/m = (13 - 1)/3 = 4$

$$l = [(m - 1)n + 1]/m = [(3 - 1)13 + 1]/3 = 9$$

(ii) $n = mi + 1 = 3 \times 4 + 1 = 13$

$$l = (m - 1)i + 1 = (3 - 1)4 + 1 = 9$$

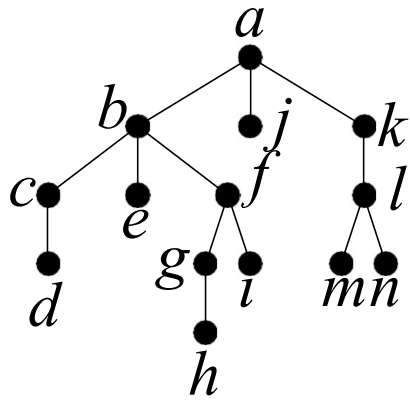
(iii) $n = (ml - 1)/(m - 1) = (3 \times 9 - 1)/(3 - 1) = 13$

$$i = (l - 1)/(m - 1) = (9 - 1)/(3 - 1) = 4$$

The **level** of a vertex v in a rooted tree is the length of the unique path from the root to this vertex. The level of the root is defined to be zero.

The **height** of a rooted tree is the maximum of the levels of vertices. In other words, the height of a rooted tree is the length of the longest path from the root to any vertex.

Example: Find the level of each vertex in the rooted tree shown in Figure 13. What is the height of this tree?



Solution:

The root a is at level 0.

Vertices b, j , and k are at level 1.

Vertices c, e, f , and l are at level 2.

Vertices d, g, i, m , and n are at level 3.

Finally, vertex h is at level 4.

Since the largest level of any vertex is 4, this tree has height 4.

Figure: A Rooted Tree.

A rooted m -ary tree of height h is **balanced** if all leaves are at levels h or $h - 1$.

Example: Which of the rooted trees shown in Figure 14 are balanced?

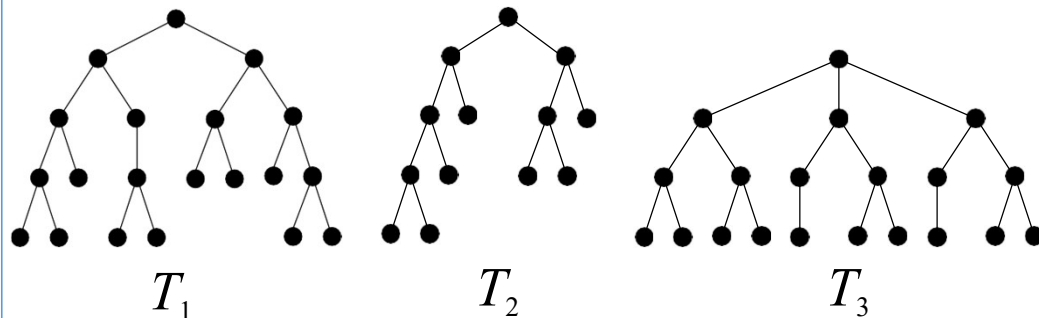


Figure: Some Rooted Trees.

Solution:

T_1 is balanced, since all its leaves are at levels 3 and 4.

T_2 is not balanced, since it has leaves at levels 2, 3, and 4.

T_3 is balanced, since all its leaves are at level 3.